decoupling, defined 75
deep learning, defined 13
deployment 268, 291–301
  alternative 299, 300
  authentication and authorization 275–285
  compliance and data protection 288–290
  example deployment to Vercel 294–299
  options for 291–292
  production checklist 292–294
  rate limiting 280–285
  secrets management 285
Docker 291
documents
  indexing 219–221
  ingestion 179
  preparing and storing for retrieval 179–185
  routes 324
  types 203
document summarization 198–216
  additional considerations for summarizing documents 212–216
  applying summarization chain and displaying to user 204
  architecture and workflow 198–202
  building 202–205
  caveats and limitations 205–208
  demonstration of app 208–211
  loading and preprocessing documents 203
  measuring cost of multiple LLM calls 207
  parsing and cleaning up documents 205
  project requirements 198
  splitting documents into manageable chunks 204
document summarization and RAG with LangChain.js, RAG web application 216–230
  adding grounding support 225–230
  key architectural components 217
  project requirements 216
  system components 219–223
  technical architecture overview 218
  web app demonstration 223
dotenv package 47
dynamic
  response generation for customer feedback 5
  updates 139

E

Elastic Stack (Elasticsearch, Logstash, Kibana) 250
embedding generation 179
embeddings 130
  IT support knowledge base 161–164
  machine language vs. 157
  restaurant menu analogy 154
  storing and retrieving with 160
  storing in database 158
  tied to particular model 157
  Vercel AI SDK 158–161
  visual representation of 156
encrypted storage 326
error boundary component 235
error handling 38, 41
Euclidean distance 161

F

factory method pattern 171
FakeLLM class 264, 266, 267
FakeRetriever class 264, 266, 267
FallbackModel class 249
feature extraction, defined 8
feedback page 312
fetch API 52
few-shot learning 146–151
  examples of 147–149
  general methodology for creating prompts 150
file-based routing 45
file uploader 84
fine-tuning 146
frontend, generative AI web applications 28
function calling 114

G

GANs (generative adversarial networks) 12
GDPR (General Data Protection Regulation) 16, 288
Gemini, vision queries 83–86
Gemini API key 44
generated content 332
generative AI 3
  AI tools and ecosystem 9
  capabilities of 4
  concerns and implications of 15–17
  generative vs. traditional AI 13–15